

AI Movie Recommendation System ■

A simple Python-based movie recommendation system that recommends movies according to the user's favorite genres.

■ Project Overview

This project is a basic AI-style recommendation system developed in Python. The user enters their favorite movie genres, and the system compares those preferences with the genres associated with available movies. Movies are then given a matching score and displayed in order of relevance.

■ Objective

The main objective of this project is to demonstrate how a simple content-based recommendation approach can be implemented using user preferences and matching scores.

■ Features

- Takes multiple favorite genres from the user
- Accepts genres separated by commas
- Converts input to lowercase for easier matching
- Compares user preferences with movie genres
- Calculates a match score for each movie
- Sorts recommendations by matching score
- Displays recommended movies
- Shows a message when no matching movie is found

■■ Technologies Used

- Python 3
- Lists
- Dictionaries
- Loops
- Conditional statements
- Lambda function
- String processing

■ How the System Works

The system first asks the user to enter favorite genres, for example **action**, **sci-fi**. The input is split into individual genres and converted to lowercase. The program then checks every movie and counts how many of its genres match the user's preferences. Movies with a higher score are placed higher in the recommendation list.

■ Recommendation Process

1. Take user genre preferences.
2. Convert the input into a list of genres.
3. Compare each preference with every movie's genres.
4. Increase the score whenever a genre matches.
5. Add movies with a score greater than zero to the recommendations.

6. Sort recommendations from highest score to lowest score.
7. Display the recommended movies and their match scores.

■ Important Code

```
user_input = input(
    "\nEnter your favorite genres (e.g. action, sci-fi): "
)

preferences = [
    genre.strip().lower()
    for genre in user_input.split(",")
]

recommendations = []

for movie, genres in movies.items():
    score = 0
    for preference in preferences:
        if preference in genres:
            score += 1
    if score > 0:
        recommendations.append((movie, score))

recommendations.sort(key=lambda x: x[1], reverse=True)
```

■■ How to Run

1. Install Python 3.
2. Open the project in VS Code.
3. Save the program, for example as **recommendation.py**.
4. Open the terminal.
5. Run the program using:

```
python recommendation.py
```

■ Example

```
Enter your favorite genres (e.g. action, sci-fi): action, sci-fi
```

```
Recommended Movies:
```

- Movie A (Match Score: 2)
- Movie B (Match Score: 1)

```
Thank you for using the AI Recommendation System!
```

■ Matching Score

The match score represents the number of user-selected genres that match the genres of a movie. For example, if a movie matches two of the user's favorite genres, its match score will be 2.

■■ Note

This is a simple rule-based/content-matching recommendation system. It does not use a trained machine learning model. The recommendations depend on the movie data and genre matching rules defined in the program.

■ Future Improvements

- Add a larger movie dataset
- Include movie ratings and popularity
- Add user rating history
- Use machine learning for personalized recommendations
- Create a graphical user interface
- Connect the system to a movie database or API
- Add collaborative filtering

■■■ Author

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■ A beginner-friendly AI-style movie recommendation project built with Python.